



## Pure Mathematics 2 Practice Paper J10 **MARK SCHEME**

### Question 1

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                            | Marks                  |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1.              | $1 + \cos Q - 3 \cos^2 Q$ $\approx 1 + \left(1 - \frac{Q^2}{2}\right) - 3\left(1 - \frac{Q}{2}\right)^2$ $\approx 1 + 1 - \frac{Q^2}{2} - 3 + 3Q^2 - \frac{3}{4}Q^4$ <p>Since <math>Q</math> is small, we can neglect the higher order terms</p> <p>So <math>1 + 1 - \frac{Q^2}{2} - 3 + 3Q^2 - \frac{3}{4}Q^4 \approx -1 + \frac{5}{2}Q^2</math> as required</p> | M1<br><br>B1<br><br>B1 |

## Question 2

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Marks                        |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------|
| (a)             | <div> <p><b>Either</b> <math>\frac{\sin(\hat{ACB})}{5} = \frac{\sin 0.6}{4}</math><br/> <math>\therefore \hat{ACB} = \arcsin(0.7058\dots)</math><br/> <math>= [0.7835\dots \text{ or } 2.358]</math><br/>           Use angles of triangle<br/> <math>\hat{ABC} = \pi - 0.6 - \hat{ACB}</math><br/>           (But as <math>AC</math> is the longest side so)<br/> <math>\hat{ABC} = 1.76</math> (*) (3sf) [Allow <math>100.7^\circ \rightarrow 1.76</math>]<br/>           In degrees <math>0.6 = 34.377^\circ</math>, <math>\hat{ACB} = 44.9^\circ</math></p> </div> <div> <p><b>or</b> <math>4^2 = b^2 + 5^2 - 2 \times b \times 5 \cos 0.6</math><br/> <math>\therefore b = \frac{10 \cos 0.6 \pm \sqrt{(100 \cos^2 0.6 - 36)}}{2}</math><br/> <math>= [6.96 \text{ or } 1.29]</math><br/>           Use sine / cosine rule with value for <math>b</math><br/> <math>\sin B = \frac{\sin 0.6}{4} \times b</math> or <math>\cos B = \frac{25 + 16 - b^2}{40}</math><br/>           (But as <math>AC</math> is the longest side so)<br/> <math>\hat{ABC} = 1.76</math> (*) (3sf)</p> </div>                                                                                         | M1<br>M1<br>M1,<br>A1<br>(4) |
| (b)             | <p><math>\left[ \hat{CBD} = \pi - 1.76 = 1.38 \right]</math> Sector area <math>= \frac{1}{2} \times 4^2 \times (\pi - 1.76) = [11.0 \sim 11.1]</math> <math>\frac{1}{2} \times 4^2 \times 79.3</math> is M0</p> <p>Area of <math>\triangle ABC = \frac{1}{2} \times 5 \times 4 \times \sin(1.76) = [9.8]</math> or <math>\frac{1}{2} \times 5 \times 4 \times \sin 101</math></p> <p>Required area = awrt 20.8 or 20.9 or 21.0 or gives 21 (2sf) after correct work.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | M1<br>M1<br>A1 (3)<br>[7]    |
| (a)             | <p>1<sup>st</sup> M1 for correct use of sine rule to find <math>\hat{ACB}</math> or cosine rule to find <math>b</math> (M0 for ABC here or for use of <math>\sin x</math> where <math>x</math> could be <math>\hat{ABC}</math>)<br/>           2<sup>nd</sup> M1 for a correct expression for angle <math>\hat{ACB}</math> (This mark may be implied by .7835 or by <math>\arcsin(.7058)</math>) and needs accuracy. In second method this M1 is for correct expression for <math>b</math> – may be implied by 6.96. [Note <math>10 \cos 0.6 \approx 8.3</math>]<br/>           (do not need two answers)<br/>           3<sup>rd</sup> M1 for a correct method to get angle <math>\hat{ABC}</math> in method (i) or <math>\sin \hat{ABC}</math> or <math>\cos \hat{ABC}</math>, in method (ii) (If <math>\sin B &gt; 1</math>, can have M1A0)<br/>           A1cso for correct work leading to 1.76 3sf. Do not need to see angle 0.1835 considered and rejected.</p> <p>1<sup>st</sup> M1 for a correct expression for sector area or a value in the range 11.0 – 11.1<br/>           2<sup>nd</sup> M1 for a correct expression for the area of the triangle or a value of 9.8</p> |                              |
| (b)             | <p>Ignore 0.31 (working in degrees) as subsequent work.<br/>           A1 for answers which round to 20.8 or 20.9 or 21.0. No need to see units.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                              |
| (a)             | <p><u>Special case</u> If answer 1.76 is assumed then usual mark is M0 M0 M0 A0. A Fully checked method may be worth M1 M1 M0 A0. A maximum of 2 marks. The mark is either 2 or 0.</p> <p><u>Either</u> M1 for <math>\hat{ACB}</math> is found to be 0.7816 (angles of triangle) then</p> <p>M1 for checking <math>\frac{\sin(\hat{ACB})}{5} = \frac{\sin 0.6}{4}</math> with conclusion giving numerical answers</p> <p>This gives a maximum mark of 2/4</p> <p><u>OR</u> M1 for <math>b</math> is found to be 6.97 (cosine rule)</p> <p>M1 for checking <math>\frac{\sin(\hat{ABC})}{b} = \frac{\sin 0.6}{4}</math> with conclusion giving numerical answers</p> <p>This gives a maximum mark of 2/4</p> <p>Candidates making this assumption need a complete method. They cannot earn M1M0.<br/>           So the score will be 0 or 2 for part (a). Circular arguments earn 0/4.</p>                                                                                                                                                                                                                                                                                              |                              |

### Question 3

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                             | Marks                  |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| (a)             | $18000 \times (0.8)^3 = £9216$ * [may see $\frac{4}{5}$ or 80% or equivalent].                                                                                                                                                                                                                                                                                                                     | B1cso (1)              |
| (b)             | $18000 \times (0.8)^n < 1000$<br>$n \log(0.8) < \log\left(\frac{1}{18}\right)$<br>$n > \frac{\log\left(\frac{1}{18}\right)}{\log(0.8)} = 12.952....$ so $n = 13$ .                                                                                                                                                                                                                                 | M1<br>M1<br>A1 cso (3) |
| (c)             | $u_5 = 200 \times (1.12)^4$ , = £314.70 or £314.71                                                                                                                                                                                                                                                                                                                                                 | M1, A1 (2)             |
| (d)             | $S_{15} = \frac{200(1.12^{15} - 1)}{1.12 - 1}$ or $\frac{200(1 - 1.12^{15})}{1 - 1.12}$ , = 7455.94..... awrt £7460                                                                                                                                                                                                                                                                                | M1A1, A1 (3)<br>[9]    |
| (a)             | B1 NB Answer is printed so need working. May see as above or $\times 0.8$ in three steps giving 14400, 11520, 9216. Do not need to see £ sign but should see 9216.                                                                                                                                                                                                                                 |                        |
| (b)             | 1 <sup>st</sup> M1 for an attempt to use $n$ th term and 1000. Allow $n$ or $n - 1$ and allow $>$ or $=$<br>2 <sup>nd</sup> M1 for use of logs to find $n$ Allow $n$ or $n - 1$ and allow $>$ or $=$<br>A1 Need $n = 13$ This is an accuracy mark and must follow award of both M marks but should not follow incorrect work using $n - 1$ for example.<br>Condone slips in inequality signs here. |                        |
| (c)             | M1 for use of their $a$ and $r$ in formula for 5 <sup>th</sup> term of GP<br>A1 cao need one of these answers – answer can imply method here<br>NB 314.7 – A0                                                                                                                                                                                                                                      |                        |
| (d)             | M1 for use of sum to 15 terms of GP using their $a$ and their $r$ (allow if formula stated correctly and one error in substitution, but must use $n$ not $n - 1$ )<br>1 <sup>st</sup> A1 for a fully correct expression (not evaluated)                                                                                                                                                            |                        |
| (b)             | Alternative Methods<br>Trial and Improvement<br>See 989.56 (or 989 or 990) identified with 12, 13 or 14 years for first M1<br>See 1236.95 (or 1236 or 1237) identified with 11, 12 or 13 years for second M1<br>Then $n = 13$ is A1 (needs both Ms)<br>Special case $18000 \times (0.8)^n < 1000$ so $n = 13$ as $989.56 < 1000$ is M1M0A0 (not discounted $n = 12$ )                              |                        |
| (c)             | May see the terms 224, 250.88, 280.99, 314.71 with a small slip for M1 A0, or done accurately for M1A1                                                                                                                                                                                                                                                                                             |                        |
| (d)             | Adds 15 terms $200 + 224 + 250.88 + \dots + (977.42)$ M1<br>Seeing 977... is A1<br>Obtains answer 7455.94 A1 or awrt £7460 NOT 7450                                                                                                                                                                                                                                                                |                        |

# Question 4

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Marks                                                                            |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                 | <p>(a) <math>(1-8x)^{\frac{1}{2}} = 1 + \left(\frac{1}{2}\right)(-8x) + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)}{2}(-8x)^2 + \frac{\left(\frac{1}{2}\right)\left(-\frac{1}{2}\right)\left(-\frac{3}{2}\right)}{3!}(-8x)^3 + \dots</math><br/> <math>= 1 - 4x - 8x^2 - 32x^3 - \dots</math></p> <p>(b) <math>\sqrt{1-8x} = \sqrt{1 - \frac{8}{100}}</math><br/> <math>= \sqrt{\frac{92}{100}} = \sqrt{\frac{23}{25}} = \frac{\sqrt{23}}{5} \quad *</math></p> <p>(c) <math>1 - 4x - 8x^2 - 32x^3 = 1 - 4(0.01) - 8(0.01)^2 - 32(0.01)^3</math><br/> <math>= 1 - 0.04 - 0.0008 - 0.00032 = 0.959168</math><br/> <math>\sqrt{23} = 5 \times 0.959168</math><br/> <math>= 4.79584</math></p> | <p>M1 A1<br/>A1; A1 (4)</p> <p>M1<br/>A1 (2)</p> <p>M1<br/>A1 (3)</p> <p>[9]</p> |

# Question 5

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                     | Marks                                        |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
|                 | <p>(a) <math>-2 \sin 2x - 3 \sin 3y \frac{dy}{dx} = 0</math></p> <p><math>\frac{dy}{dx} = -\frac{2 \sin 2x}{3 \sin 3y}</math>      Accept <math>\frac{2 \sin 2x}{-3 \sin 3y}, \frac{-2 \sin 2x}{3 \sin 3y}</math></p>                                                                                                                                                                      | <p>M1 A1</p> <p>A1 (3)</p>                   |
|                 | <p>(b) At <math>x = \frac{\pi}{6}</math>, <math>\cos\left(\frac{2\pi}{6}\right) + \cos 3y = 1</math></p> <p><math>\cos 3y = \frac{1}{2}</math></p> <p><math>3y = \frac{\pi}{3} \Rightarrow y = \frac{\pi}{9}</math>      awrt 0.349</p>                                                                                                                                                    | <p>M1</p> <p>A1</p> <p>A1 (3)</p>            |
|                 | <p>(c) At <math>\left(\frac{\pi}{6}, \frac{\pi}{9}\right)</math>, <math>\frac{dy}{dx} = -\frac{2 \sin 2\left(\frac{\pi}{6}\right)}{3 \sin 3\left(\frac{\pi}{9}\right)} = -\frac{2 \sin \frac{\pi}{3}}{3 \sin \frac{\pi}{3}} = -\frac{2}{3}</math></p> <p><math>y - \frac{\pi}{9} = -\frac{2}{3}\left(x - \frac{\pi}{6}\right)</math></p> <p>Leading to <math>6x + 9y - 2\pi = 0</math></p> | <p>M1</p> <p>M1</p> <p>A1 (3)</p> <p>[9]</p> |



# Question 6

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (a)             | $y = \sec x = \frac{1}{\cos x} = (\cos x)^{-1}$ $\frac{dy}{dx} = -1(\cos x)^{-2}(-\sin x)$ $\frac{dy}{dx} = \left\{ \frac{\sin x}{\cos^2 x} \right\} = \left( \frac{1}{\cos x} \right) \left( \frac{\sin x}{\cos x} \right) = \underline{\underline{\sec x \tan x}}$                                                                                                                                                                                                                                               | $\frac{dy}{dx} = \pm ((\cos x)^{-2}(\sin x))$ $-1(\cos x)^{-2}(-\sin x) \text{ or } (\cos x)^{-2}(\sin x)$ <p>Convincing proof.<br/>Must see both <u>underlined steps</u>.</p> <p>M1<br/>A1<br/>A1 AG<br/>(3)</p>                                                                                                                                                                                                                                                                                               |
| (b)             | $y = e^{2x} \sec 3x$ $\left\{ \begin{array}{ll} u = e^{2x} & v = \sec 3x \\ \frac{du}{dx} = 2e^{2x} & \frac{dv}{dx} = 3 \sec 3x \tan 3x \end{array} \right\}$ $\frac{dy}{dx} = 2e^{2x} \sec 3x + 3e^{2x} \sec 3x \tan 3x$                                                                                                                                                                                                                                                                                          | <div style="border: 1px solid black; padding: 5px; display: inline-block;">Seen<br/>or implied</div> <p>Either <math>e^{2x} \rightarrow 2e^{2x}</math> or<br/><math>\sec 3x \rightarrow 3 \sec 3x \tan 3x</math><br/>Both <math>e^{2x} \rightarrow 2e^{2x}</math> and<br/><math>\sec 3x \rightarrow 3 \sec 3x \tan 3x</math></p> <p>Applies <math>vu' + uv'</math> correctly for their<br/><math>u, u', v, v'</math></p> $2e^{2x} \sec 3x + 3e^{2x} \sec 3x \tan 3x$ <p>M1<br/>A1<br/>A1<br/>A1 isw<br/>(4)</p> |
| (c)             | <p>Turning point <math>\Rightarrow \frac{dy}{dx} = 0</math></p> <p>Hence, <math>e^{2x} \sec 3x (2 + 3 \tan 3x) = 0</math></p> <p>{Note <math>e^{2x} \neq 0</math>, <math>\sec 3x \neq 0</math>, so <math>2 + 3 \tan 3x = 0</math>,}</p> <p>giving <math>\tan 3x = -\frac{2}{3}</math></p> <p><math>\Rightarrow 3x = -0.58800 \Rightarrow x = \{a\} = -0.19600\dots</math></p> <p>Hence, <math>y = \{b\} = e^{2(-0.196)} \sec(3 \times -0.196)</math></p> <p><math>= 0.812093\dots = 0.812 \text{ (3sf)}</math></p> | <p>Sets their <math>\frac{dy}{dx} = 0</math> and factorises (or cancels)<br/>out at least <math>e^{2x}</math> from at least two terms.</p> <p><math>\tan 3x = \pm k; k \neq 0</math></p> <p>Either awrt <math>-0.196^\circ</math> or awrt <math>-11.2^\circ</math></p> <p>M1<br/>A1<br/>M1<br/>A1<br/>A1<br/>A1<br/>0.812 A1 cao<br/>(4)</p>                                                                                                                                                                    |

[11]

**Part (c):** If there are any EXTRA solutions for  $x$  (or  $a$ ) inside the range  $-\frac{\pi}{6} < x < \frac{\pi}{6}$ , ie.  $-0.524 < x < 0.524$  or ANY EXTRA solutions for  $y$  (or  $b$ ), (for these values of  $x$ ) then withhold the final accuracy mark.  
Also ignore EXTRA solutions outside the range  $-\frac{\pi}{6} < x < \frac{\pi}{6}$ , ie.  $-0.524 < x < 0.524$ .





## Question 8

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                               | Marks                                                                                 |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
|                 | $\frac{dA}{dt} = 1.5$ $A = \pi r^2 \Rightarrow \frac{dA}{dr} = 2\pi r$ <p>When <math>A = 2</math></p> $2 = \pi r^2 \Rightarrow r = \sqrt{\frac{2}{\pi}} (= 0.797\ 884 \dots)$ $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$ $1.5 = 2\pi r \frac{dr}{dt}$ $\frac{dr}{dt} = \frac{1.5}{2\pi\sqrt{\frac{2}{\pi}}} \approx 0.299$ | <p>B1</p> <p>B1</p> <p>M1</p> <p>M1</p> <p>awrt 0.299</p> <p>A1</p> <p><b>[5]</b></p> |



# Question 9

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| (i)             | $y = \frac{\ln(x^2 + 1)}{x}$ $u = \ln(x^2 + 1) \Rightarrow \frac{du}{dx} = \frac{2x}{x^2 + 1}$ <p>Apply quotient rule: <math>\left\{ \begin{array}{l} u = \ln(x^2 + 1) \quad v = x \\ \frac{du}{dx} = \frac{2x}{x^2 + 1} \quad \frac{dv}{dx} = 1 \end{array} \right\}</math></p> $\frac{dy}{dx} = \frac{\left( \frac{2x}{x^2 + 1} \right)(x) - \ln(x^2 + 1)}{x^2}$ $\left\{ \frac{dy}{dx} = \frac{2}{(x^2 + 1)} - \frac{1}{x^2} \ln(x^2 + 1) \right\}$ | <p><math>\ln(x^2 + 1) \rightarrow \frac{\text{something}}{x^2 + 1}</math> M1</p> <p><math>\ln(x^2 + 1) \rightarrow \frac{2x}{x^2 + 1}</math> A1</p> <p>Applying <math>\frac{xu' - \ln(x^2 + 1)v'}{x^2}</math> correctly. M1</p> <p>Correct differentiation with correct bracketing but allow recovery. A1</p> <p>{Ignore subsequent working.}</p> <p>(4)</p>                                                                                                                                                                                                        |
| (ii)            | $x = \tan y$ $\frac{dx}{dy} = \sec^2 y$ $\frac{dy}{dx} = \frac{1}{\sec^2 y} \quad \{= \cos^2 y\}$ $\frac{dy}{dx} = \frac{1}{1 + \tan^2 y}$ <p>Hence, <math>\frac{dy}{dx} = \frac{1}{1 + x^2}</math>, (as required)</p>                                                                                                                                                                                                                                 | <p><math>\tan y \rightarrow \sec^2 y</math> or an attempt to differentiate <math>\frac{\sin y}{\cos y}</math> using either the quotient rule or product rule. M1*</p> <p><math>\frac{dx}{dy} = \sec^2 y</math> A1</p> <p>Finding <math>\frac{dy}{dx}</math> by reciprocating <math>\frac{dx}{dy}</math>. dM1*</p> <p>For writing down or applying the identity <math>\sec^2 y = 1 + \tan^2 y</math>, which must be applied/stated completely in y. dM1*</p> <p>For the correct proof, leading on from the previous line of working. A1 AG</p> <p>(5)</p> <p>[9]</p> |



# Question 10

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Marks                                                                            |
|-----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                 | <p>(a) <math>\frac{dx}{du} = -2 \sin u</math></p> <p><math>\int \frac{1}{x^2 \sqrt{4-x^2}} dx = \int \frac{1}{(2 \cos u)^2 \sqrt{4-(2 \cos u)^2}} \times -2 \sin u du</math></p> <p><math>= \int \frac{-2 \sin u}{4 \cos^2 u \sqrt{4 \sin^2 u}} du</math> Use of <math>1 - \cos^2 u = \sin^2 u</math></p> <p><math>= -\frac{1}{4} \int \frac{1}{\cos^2 u} du</math> <math>\pm k \int \frac{1}{\cos^2 u} du</math></p> <p><math>= -\frac{1}{4} \tan u (+C)</math> <math>\pm k \tan u</math></p> <p><math>x = \sqrt{2} \Rightarrow \sqrt{2} = 2 \cos u \Rightarrow u = \frac{\pi}{4}</math></p> <p><math>x = 1 \Rightarrow 1 = 2 \cos u \Rightarrow u = \frac{\pi}{3}</math></p> <p><math>\left[ -\frac{1}{4} \tan u \right]_{\frac{\pi}{3}}^{\frac{\pi}{4}} = -\frac{1}{4} \left( \tan \frac{\pi}{4} - \tan \frac{\pi}{3} \right)</math></p> <p><math>= -\frac{1}{4} (1 - \sqrt{3}) \left( = \frac{\sqrt{3}-1}{4} \right)</math></p> | <p>B1</p> <p>M1</p> <p>M1</p> <p>M1</p> <p>M1</p> <p>M1</p> <p>A1</p> <p>(7)</p> |



# Question 11

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks                                                                                        |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|
|                 | (a) $\int \frac{9x+6}{x} dx = \int \left( 9 + \frac{6}{x} \right) dx$ $= 9x + 6 \ln x (+C)$                                                                                                                                                                                                                                                                                                                                                                  | M1<br>A1<br>(2)                                                                              |
|                 | (b) $\int \frac{1}{y^{\frac{1}{3}}} dy = \int \frac{9x+6}{x} dx$<br>$\int y^{-\frac{1}{3}} dy = \int \frac{9x+6}{x} dx$<br>$\frac{y^{\frac{2}{3}}}{\frac{2}{3}} = 9x + 6 \ln x (+C)$<br>$\frac{3}{2} y^{\frac{2}{3}} = 9x + 6 \ln x (+C)$<br>$y = 8, x = 1$<br>$\frac{3}{2} 8^{\frac{2}{3}} = 9 + 6 \ln 1 + C$<br>$C = -3$<br>$y^{\frac{2}{3}} = \frac{2}{3}(9x + 6 \ln x - 3)$<br>$y^2 = (6x + 4 \ln x - 2)^3 \quad \left( = 8(3x + 2 \ln x - 1)^3 \right)$ | Integral signs not necessary<br>B1<br><br>M1<br>A1ft<br><br>M1<br>A1<br><br>A1<br>(6)<br>[8] |



# Question 12

| Question Number | Scheme                                                                                                                                                                                                                                                             | Marks                                                                                                                   |
|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| (a)             | $\int x \ln x \, dx = \frac{x^2}{2} \ln x - \int \frac{x^2}{2} \times \frac{1}{x} \, dx$ $= \frac{x^2}{2} \ln x - \int \frac{x}{2} \, dx$ $= \frac{x^2}{2} \ln x - \frac{x^2}{4} (+C)$                                                                             | <div style="border-left: 1px solid black; border-right: 1px solid black; padding: 0 10px;"> M1 A1<br/><br/>M1 A1 </div> |
| (b)             | $\left[ \frac{x^2}{2} \ln x - \frac{x^2}{4} \right]_1^4 = (8 \ln 4 - 4) - \left( -\frac{1}{4} \right)$ $= 8 \ln 4 - \frac{15}{4}$ $= 8(2 \ln 2) - \frac{15}{4} \quad \ln 4 = 2 \ln 2 \text{ seen or implied}$ $= \frac{1}{4}(64 \ln 2 - 15) \quad a = 64, b = -15$ | M1<br><br>A1<br><br>(7)<br>[13]                                                                                         |



# Question 13

| Question Number | Scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
|                 | <p>(a) <math>y = 0 \Rightarrow t(9 - t^2) = t(3 - t)(3 + t) = 0</math><br/> <math>t = 0, 3, -3</math> Any one correct value B1<br/> At <math>t = 0</math>, <math>x = 5(0)^2 - 4 = -4</math> Method for finding one value of <math>x</math> M1<br/> At <math>t = 3</math>, <math>x = 5(3)^2 - 4 = 41</math><br/> (At <math>t = -3</math>, <math>x = 5(-3)^2 - 4 = 41</math>)<br/> At <math>A</math>, <math>x = -4</math>; at <math>B</math>, <math>x = 41</math> Both A1 (3)</p> <p>(b) <math>\frac{dx}{dt} = 10t</math> Seen or implied B1<br/> <math>\int y \, dx = \int y \frac{dx}{dt} \, dt = \int t(9 - t^2)10t \, dt</math> M1 A1<br/> <math>= \int (90t^2 - 10t^4) \, dt</math><br/> <math>= \frac{90t^3}{3} - \frac{10t^5}{5} (+C) \quad (= 30t^3 - 2t^5 (+C))</math> A1<br/> <math>\left[ \frac{90t^3}{3} - \frac{10t^5}{5} \right]_0^3 = 30 \times 3^3 - 2 \times 3^5 \quad (= 324)</math> M1<br/> <math>A = 2 \int y \, dx = 648 \quad (\text{units}^2)</math> A1 (6)<br/> <b>[9]</b></p> |       |